Morphology of Pt nanoparticles: An investigation from descriptors

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Contents

1. previous work on oblateness and problems

2. predictions using an alternative descriptor: surface volume ratio(using published data)

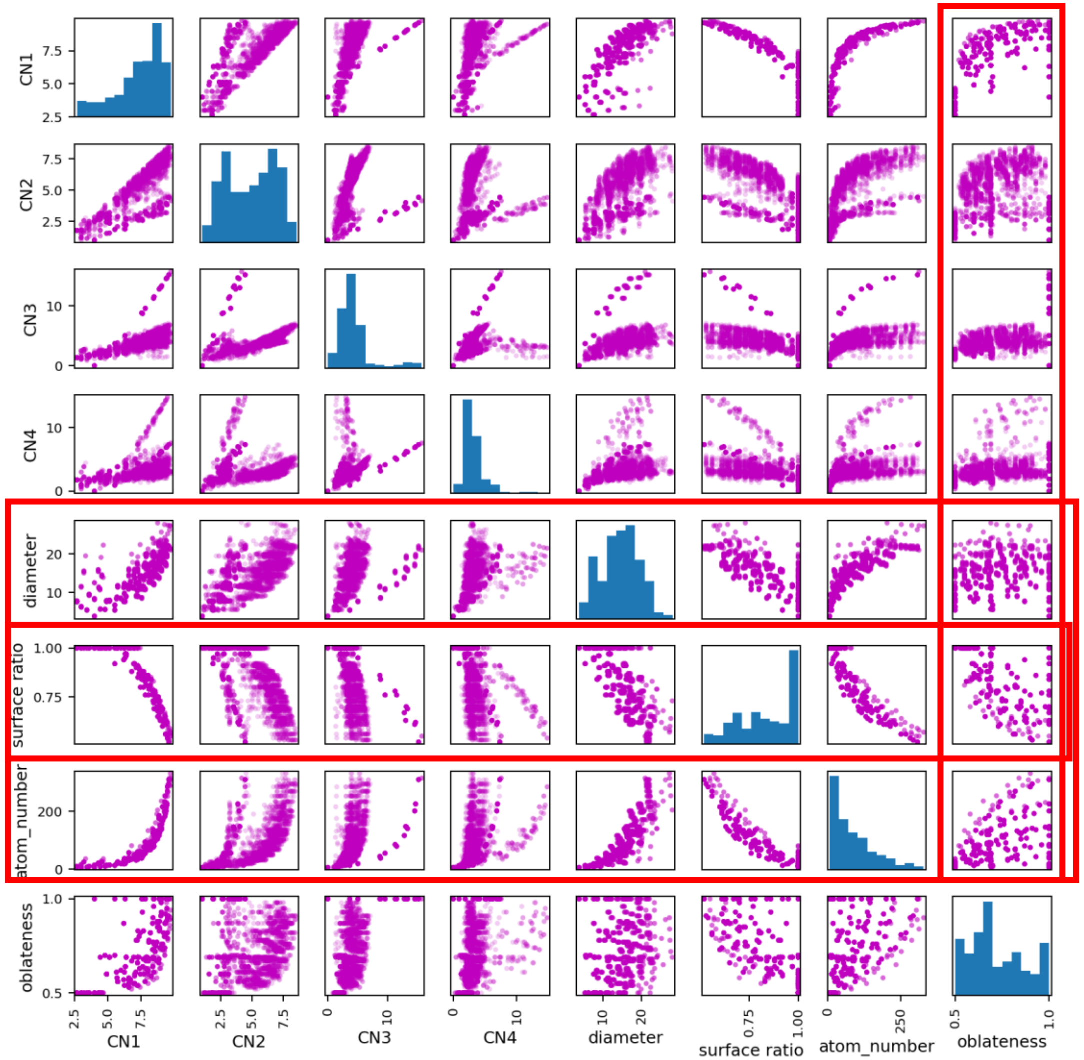
3. idea about using descriptors to reconstruct nanoparticles' image

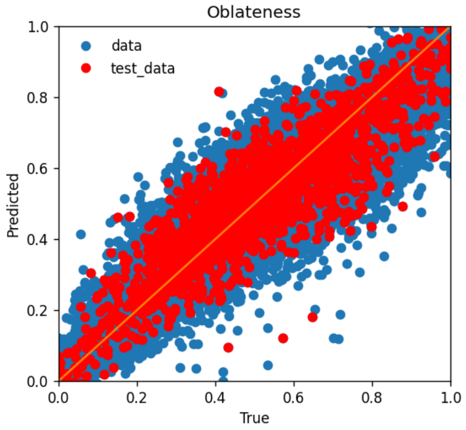
Previous NN work on shape

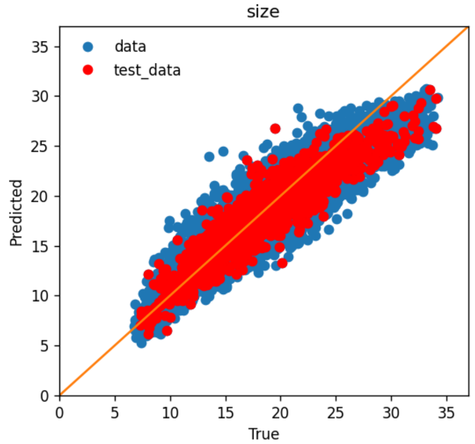
Oblateness_PCA:

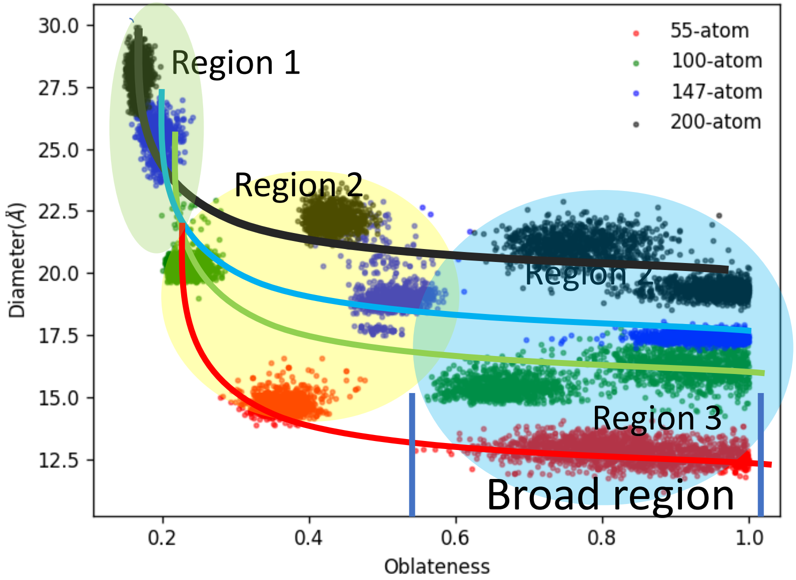
- Moments of inertia: I_0, I_1, I_2
- $a = PC1_{max} - PC1_{min}$
- $b = PC2_{max} - PC2_{min}$
- $c = PC3_{max} - PC3_{min}$
- $\eta = c/b$

XANES → Shape(oblateness)

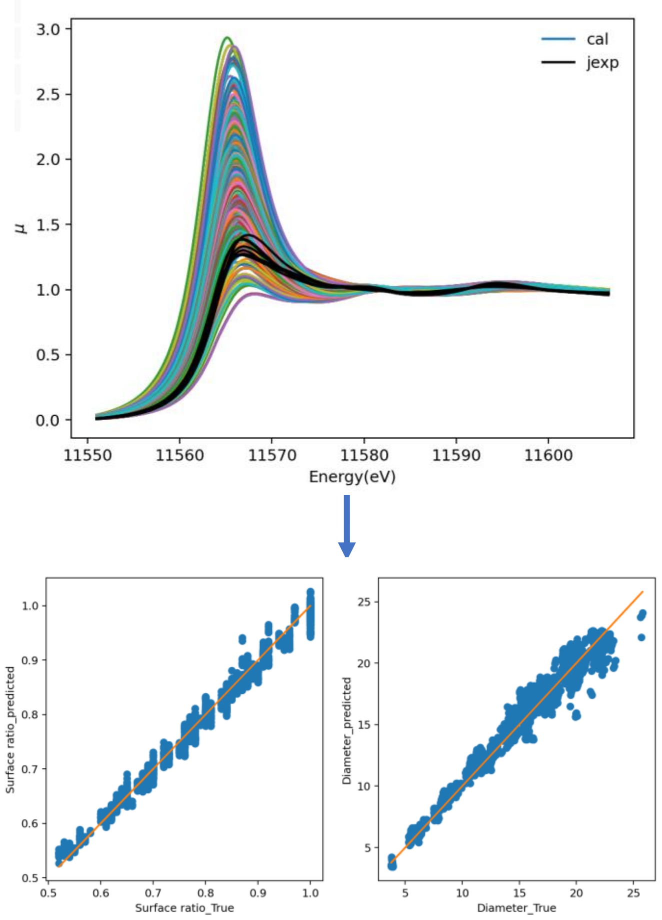








NN prediction on surface volume ratio(inverse problem)



	$d_{TEM}(nm)$[1]	CN[1]	$d_{predict}(nm)$	Surface_volume_ratio (predicted)
S3	0.9(2)	7.7(4)	1.4	0.81
S1	1.1(2)	7.4(4)	1.5	0.83
S2	1.2(3)	8.1(3)	2.1	0.63
S4	1.2(2)	8.9(3)	1.9	0.65
A1	0.9(2)	6.3(3)	1.1	0.91
A2	1.1(3)	6.6(4)	1.2	0.90

More atoms inside particle

Surface volume ratio is good descriptor!

But what is the shape of investigated particle is still unknown.

Can we reconstruct the particle or particle distribution based on several known descriptors?

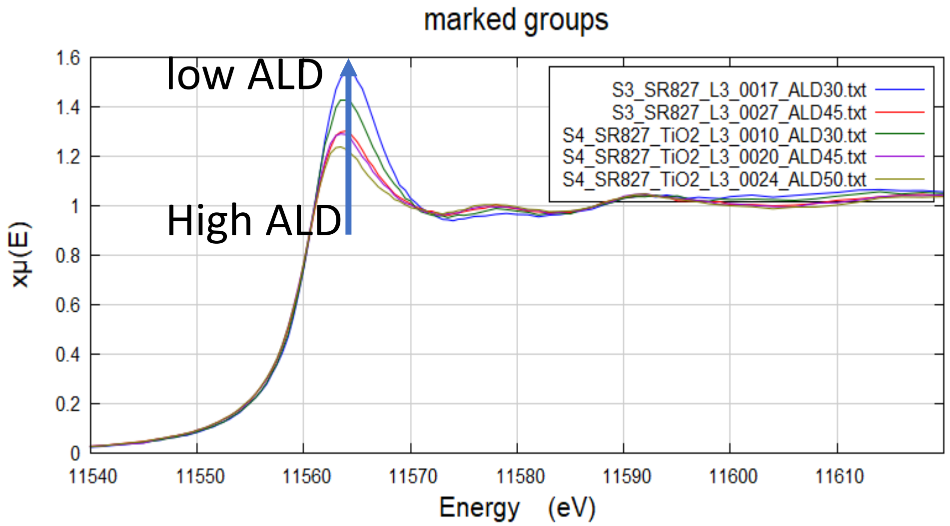
For example, can we reconstruct particles from diameter, surface volume ratio and number of atoms ?

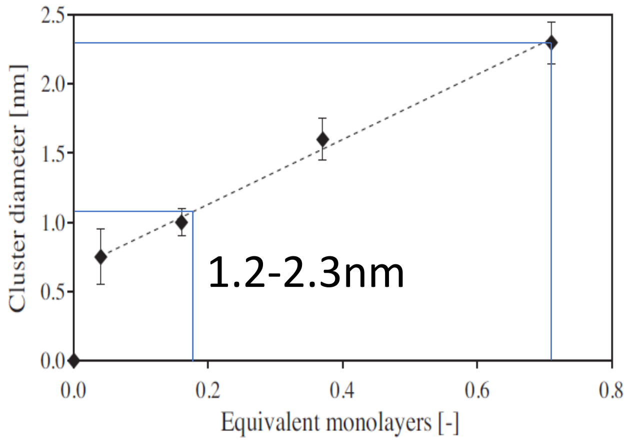
Get coordinates from descriptor!!

CNs → Shapes

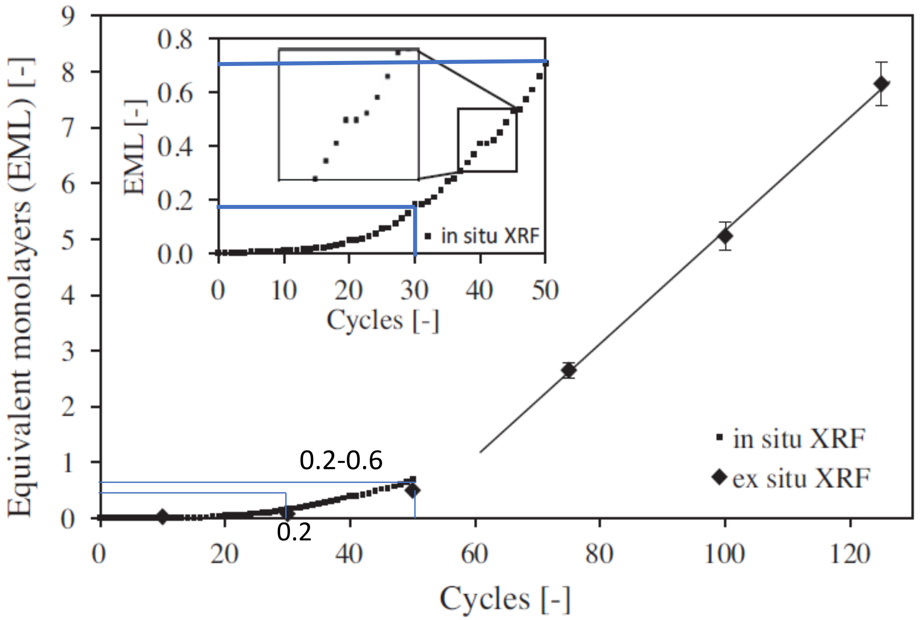
[1]. Timoshenko, Frenkel J. Phys. Chem. Lett. 2017, 8, 5091-5098

NN prediction on surface volume ratio(inverse problem)-ALD data





	$d_{published}(nm)$ [2]	$d_{predict}(nm)$	SVR (predicted)	# of atoms
S3_ALD30	1.2-2.3	2.1	0.71 More atoms on surface	131
S3_ALD45		2.3	0.62	196
S4_ALD30		1.8	0.65	149
S4_ALD45		2.4	0.66 More atoms on surface	187
S4_ALD50		2.5	0.67	183



[2]. M. Filez, H. Poelman, R.K. Ramachandran, J. Dendooven, et al. Catalysis Today 2014, 229, 2-13

Think about the connection between features and an object

list all red fruits

Here is a list of red fruits:

1. Apple (red varieties)
2. Strawberry
3. Cherry
4. Raspberry
5. Red Grape
6. Watermelon (red-fleshed varieties)
7. Pomegranate
8. Cranberry
9. Red Currant
10. Red Pear
11. Red Plum
12. Tomato (technically a fruit)
13. Red Dragon Fruit (Pitaya)
14. Blood Orange (flesh inside)
15. Red Pepper (Bell Pepper, technically a fruit)
16. Red Guava
17. Lingonberry
18. Goji Berry
19. Red Mulberry
20. Red Passion Fruit

list all red fruits that have many seeds

Here are some red fruits that have many seeds:

1. **Pomegranate** - Contains numerous seeds (arils) inside the fruit.
2. **Strawberry** - Has many small yellow seeds (achenes) on its surface.
3. **Raspberry** - Composed of multiple drupelets, each containing a seed.
4. **Red Dragon Fruit (Pitaya)** - The flesh contains numerous small black seeds.
5. **Passion Fruit (red variety)** - Filled with many small seeds surrounded by juicy pulp.
6. **Watermelon (red-fleshed varieties)** - Contains many black seeds embedded in the flesh.
7. **Red Bell Pepper** - Contains many small seeds inside the hollow fruit.

These fruits are known for their vibrant red color and their abundance of seeds, which contribute to their unique textures and flavors.

list all red fruits having many seeds on its surface

The primary red fruit that has many seeds on its surface is the **strawberry**.

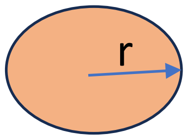
Instead of giving only one solution, GA will provide a set of optimized results

How to make a profile for a nanoparticle

Size:

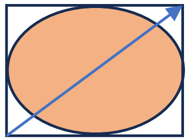
Diameter_2r:

- $\vec{r} = (x, y, z)$
- Find the mean positions($\vec{r}_{mean}$)
- $d = 2\max(|\vec{r} - \vec{r}_{mean}|)$



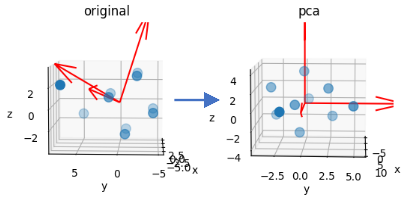
Diameter_PCA:

- PCA dimensions: $\vec{r}(x_{pc}, y_{pc}, z_{pc})$
- $d = |\vec{r}|$



Diameter_xy:

- $\vec{r} = (x, y)$
- Find the mean position($\vec{r}_{mean}$)
- $d = 2\max(|\vec{r} - \vec{r}_{mean}|)$



Oblateness_PCA:

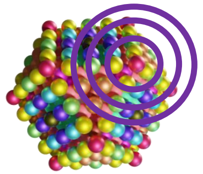
- $a = PC1_{max} - PC1_{min}$
- $b = PC2_{max} - PC2_{min}$
- $c = PC3_{max} - PC3_{min}$
- $\eta = c/b$

X-ray → Structure

Shape ↔ Size

Structure:

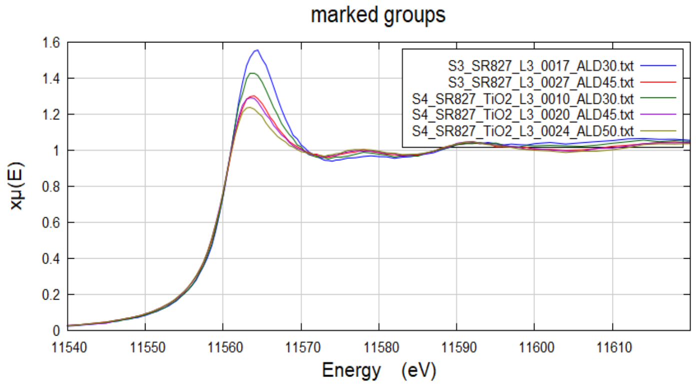
Averaged CN1, CN2, CN3, CN4



MIAD (mean interatomic distance):

- $\boldsymbol{D_{atoms}}$ is the atom-atom distance matrix of the particle
- $MIAD = \frac{1}{N(N-1)} \sum \boldsymbol{D_{atoms}}$

EXPERIMENTALLY:



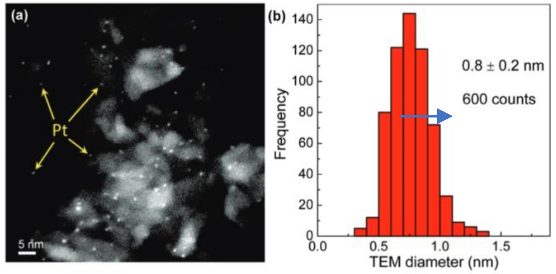
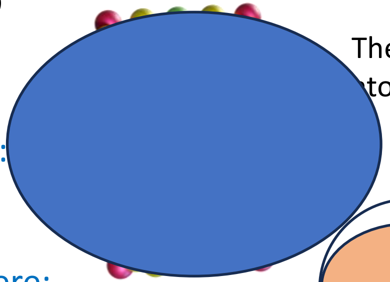


Figure 1. (a) HAADF STEM image of polymer-free micellar Pt NPs supported on γ-Al_2O_3 (S1). The image was taken after polymer removal and NP reduction in H_2 at 375 °C. (b) Histogram of the NP diameter obtained from STEM images.

Mostafa, Frenkel et al. JACS 2010, 132, 15714-15719

hape
ution

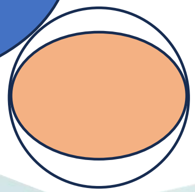
Shape:



The ratio of colorful atoms to total atoms

Surface volume ratio:

- $SVR = \frac{N_{CN<12}}{N_{total}}$

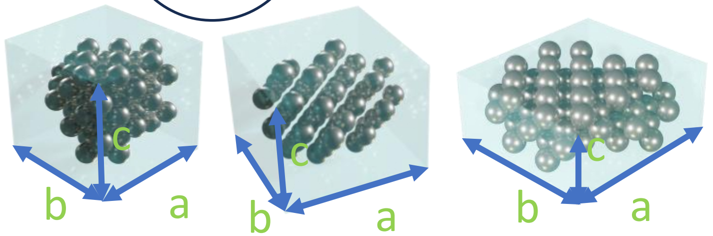


Departure from sphere:

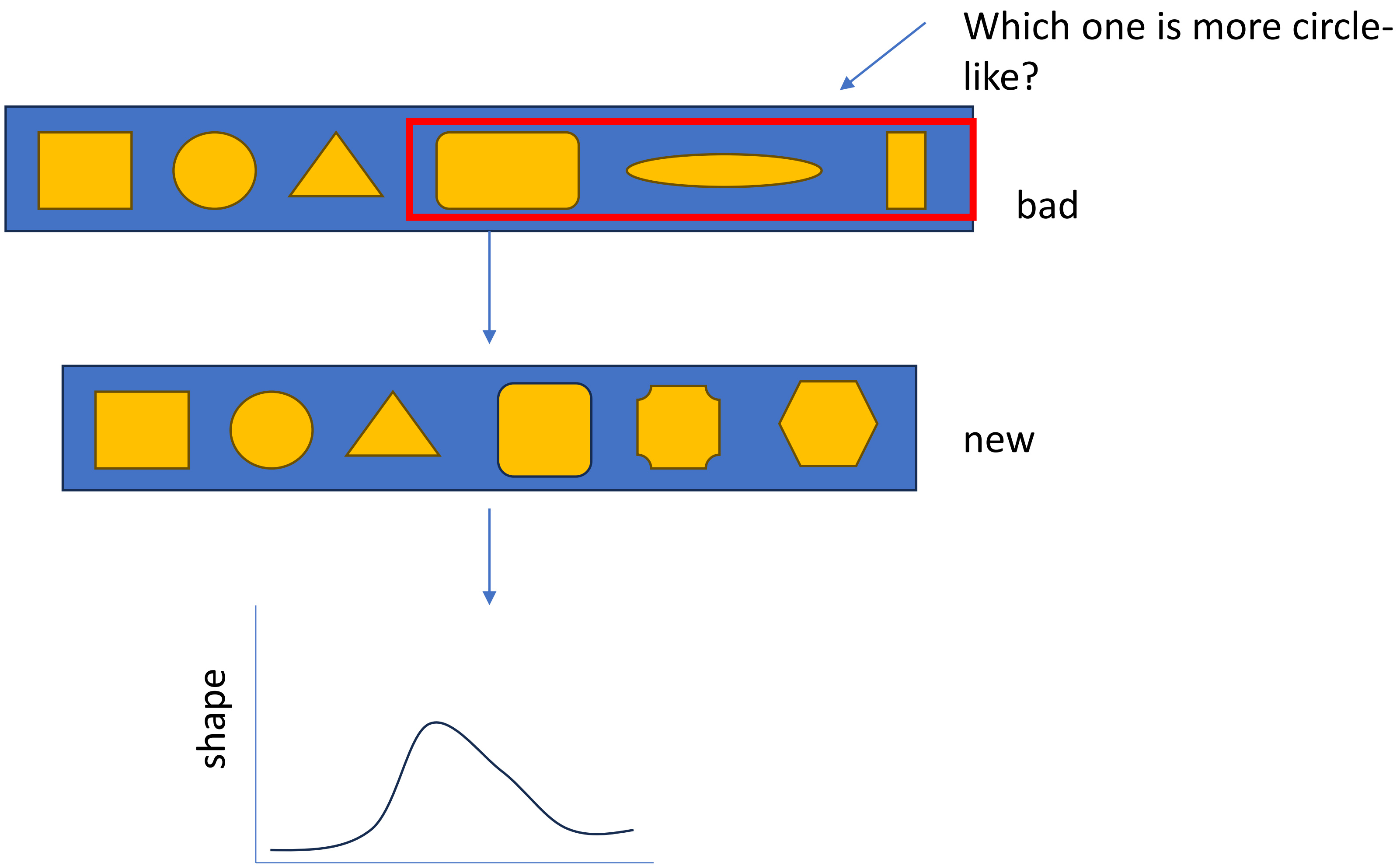
- Moments of inertia: I_0, I_1, I_2
- $\zeta = \frac{(I_2-I_1)^2+(I_1-I_0)^2+(I_0-I_2)^2}{(I_0^2+I_1^2+I_2^2)}$

Oblateness:

- Moments of inertia: I_0, I_1, I_2
- $\eta = \frac{2I_1-I_0-I_2}{I_2}$



Question:
How to solve the inverse problem for really small particle(spectrum->shape, size)


Which one is more circle-like?
bad
new
shape

Design of genetic algorithm for investigating nanoparticle morphology

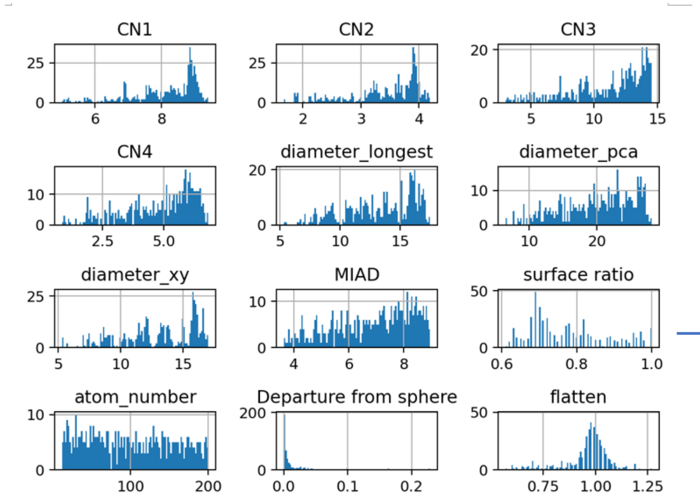
$y(P|d_s)$

Define descriptors

$(SVR, N_{atoms}, D, CN \ldots)$

Neural network prediction

Experimental observation

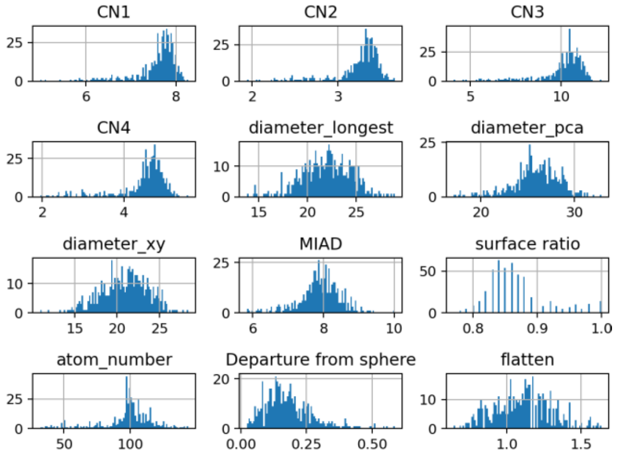


Initial state

Construct a population of particles

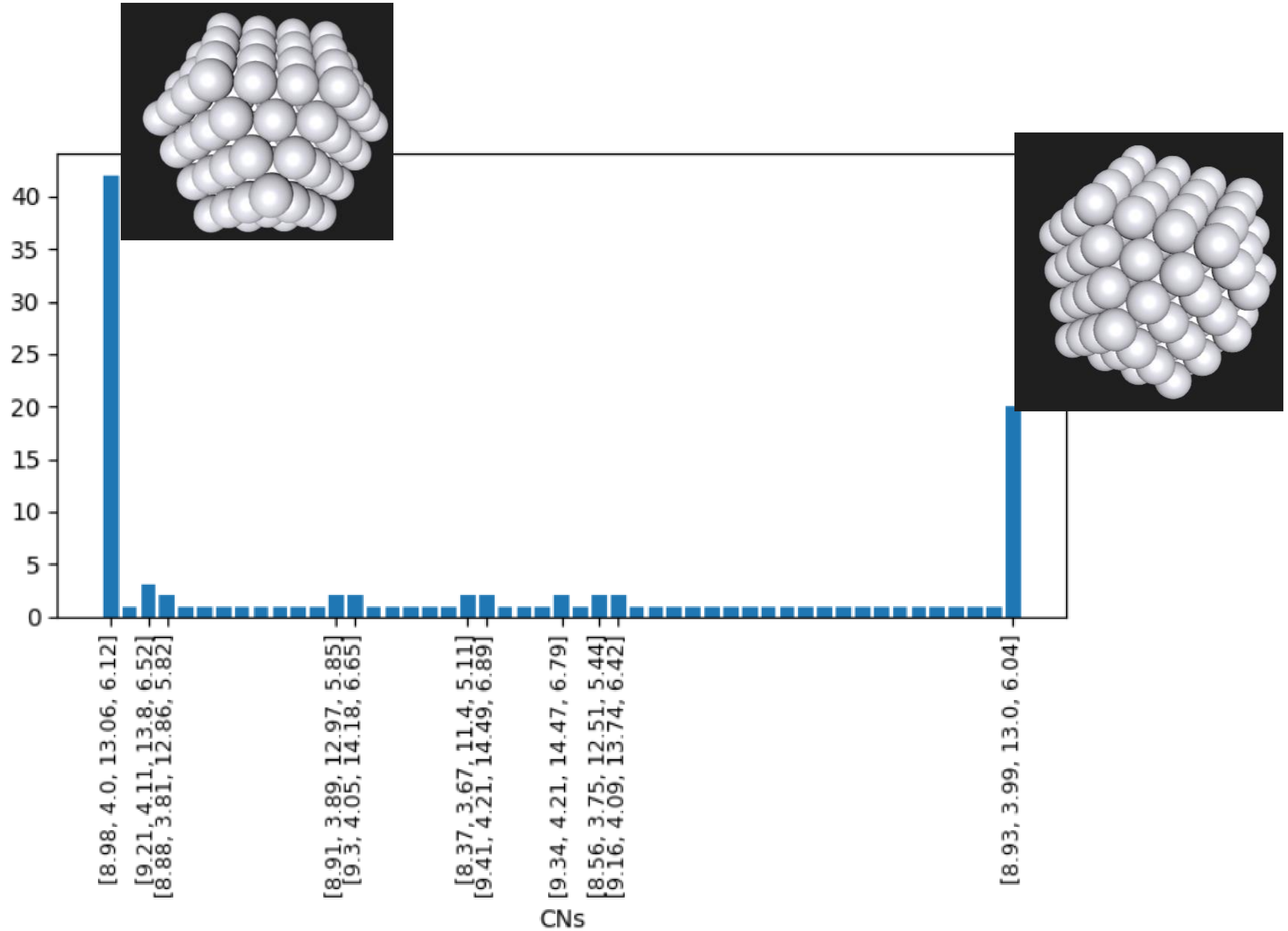
Genetic algorithm

Optimized distribution of descriptors



Example 1

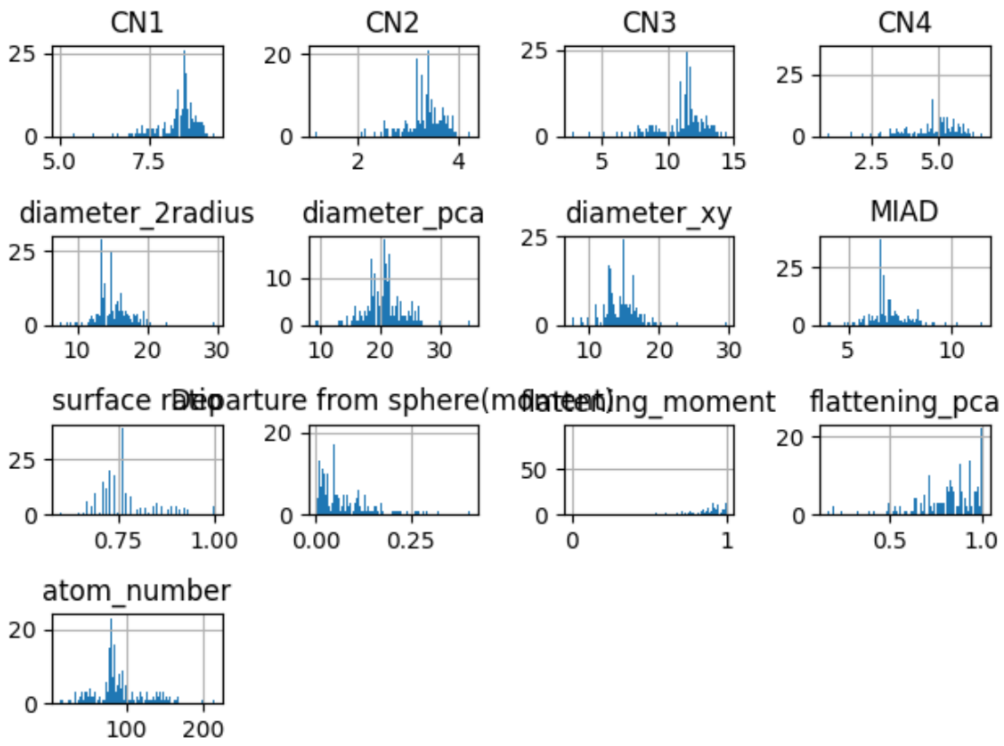
```
ini_configurations={
        "max_num_atoms":200,
        "generations":30,
        "population_size":100,
        "lc":3.77,
        "mutation_rate":0.3,
        "cross_over_rate":0.3,
        "elite_fraction":0.05,
        "descriptors":{
                "diameter_2radius":15.99,
                "surface ratio":0.63,
                "atom_number":147}
      }
```

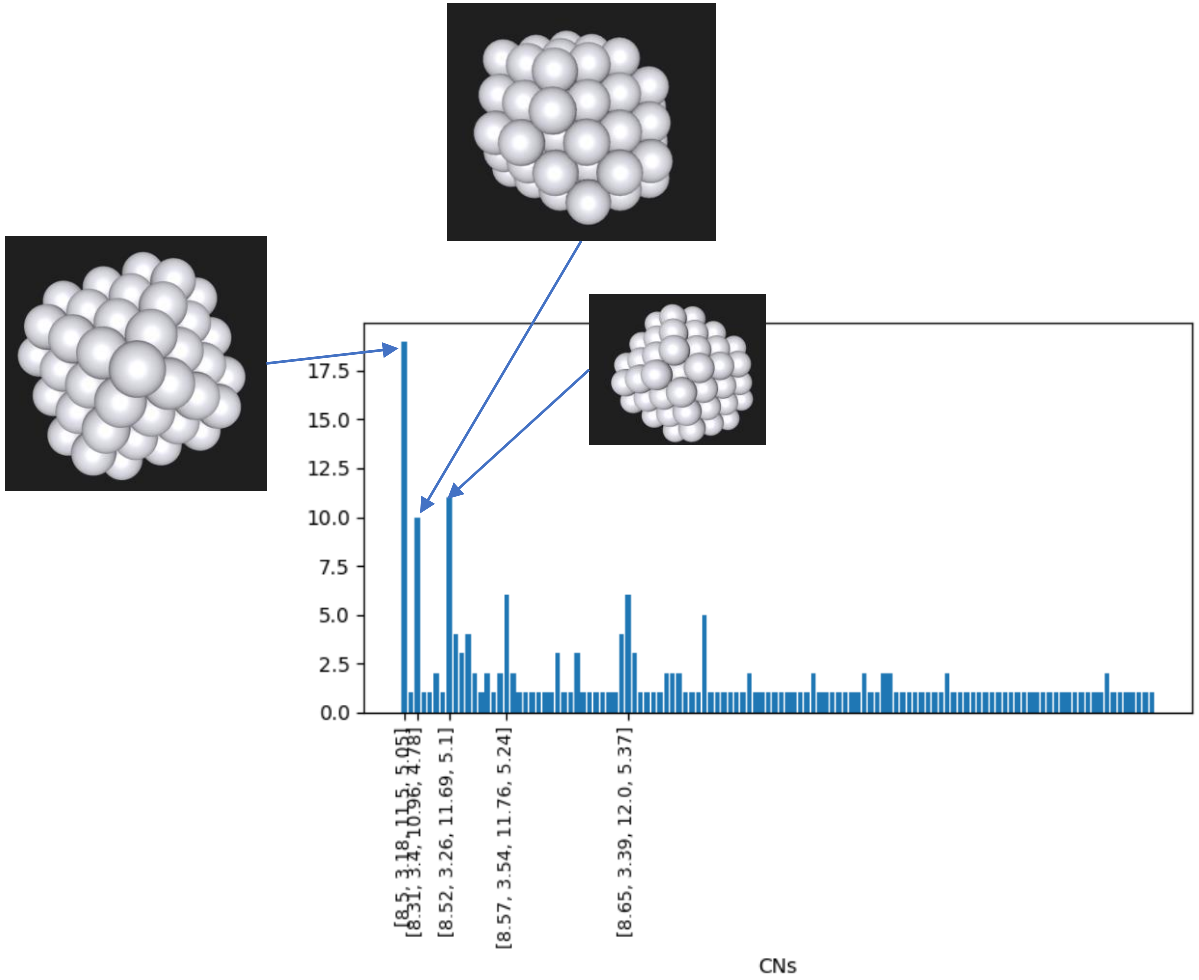


Examples 2

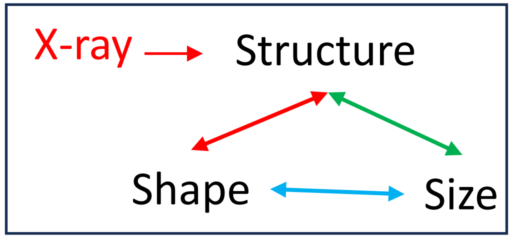
Flattening_pca=1
Atom_number=85

Final distribution





Plan

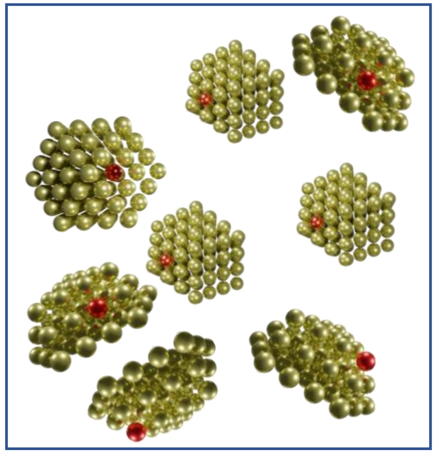


- The work goal is to understand morphology of nanoparticles.
- Surface volume ratio is a good morphological descriptor.
- Genetic algorithm maps descriptors to particle structures, which can help us to study other unknown descriptors and mophology.
- How to make a profile of nanoparticle(the minimal number of descriptors should be considered to define a particle)->strong or weak descriptors.

New shape descriptors are required for the NN work!

Supplementary materials

The inverse problem for XAS



$\{\vec{r_i}\}$ $\xrightarrow{\text{encode}}$ μ $\xrightarrow{\text{decode}}$ $\{\vec{r_i}'\}$

- Measurements
- calculations

- fitting
- Machine learning

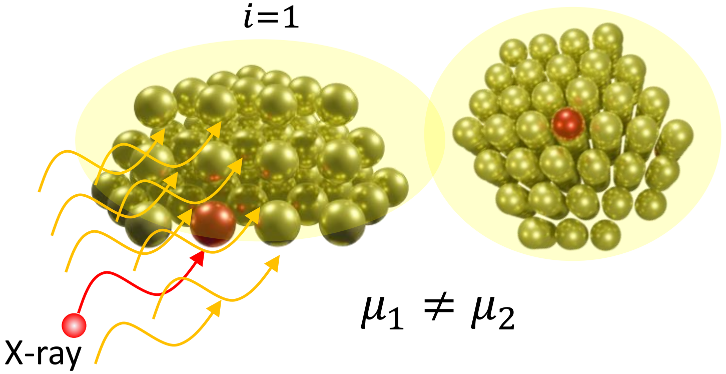
- Descriptors

$(CN, D, shape\ldots)$

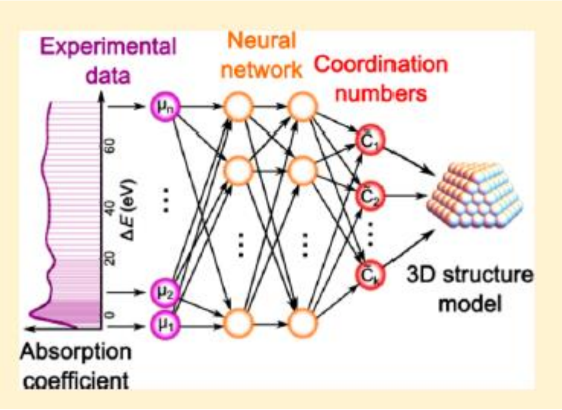
↓

morphology

$$\mu_1 = \frac{1}{N}\sum_{i=1}^{N}\mu_i \qquad \mu_2 = \frac{1}{N}\sum_{i=1}^{N}\mu_i'$$



$\mu_1 \neq \mu_2$



Timoshenko, Frenkel J. Phys. Chem. Lett. 2017, 8, 5091–5098

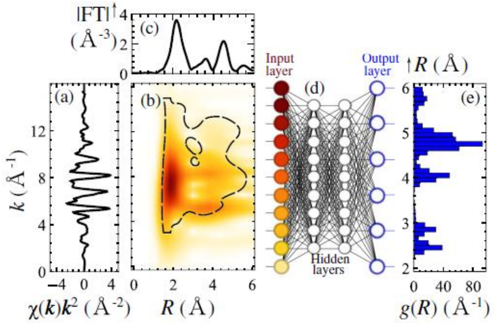
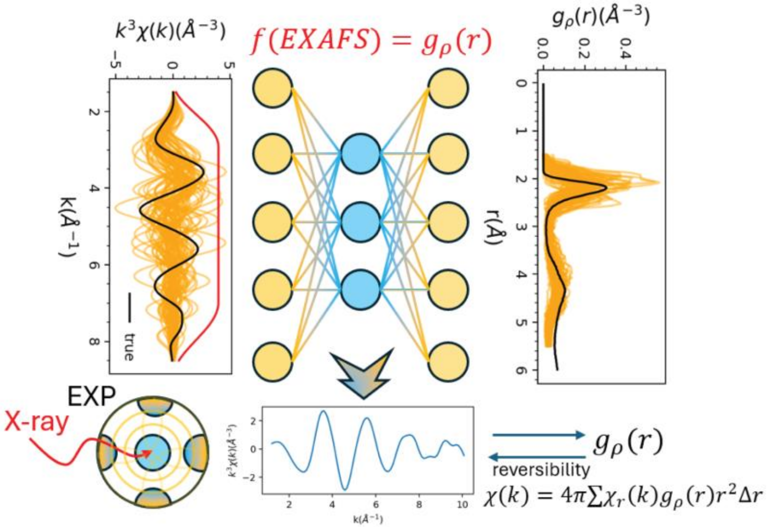


FIG. 1. Fe K-edge EXAFS for bcc iron (a). Modulus of its Morlet WT is shown in (b), while the modulus of Fourier transform (FT) is shown in (c). The dashed line in (b) indicates the region in k and R space, established as the most sensitive to structure variations. WT data are processed by NN (d), to map features in wavelet-transformed spectra to the features in RDF, approximated with a histogram (e).

Timoshenko, Frenkel PHYSICAL REVIEW LETTERS 120, 225502 (2018)



Zheng, Frenkel J. Phys. Chem. C 2024, 128, 7635–7642

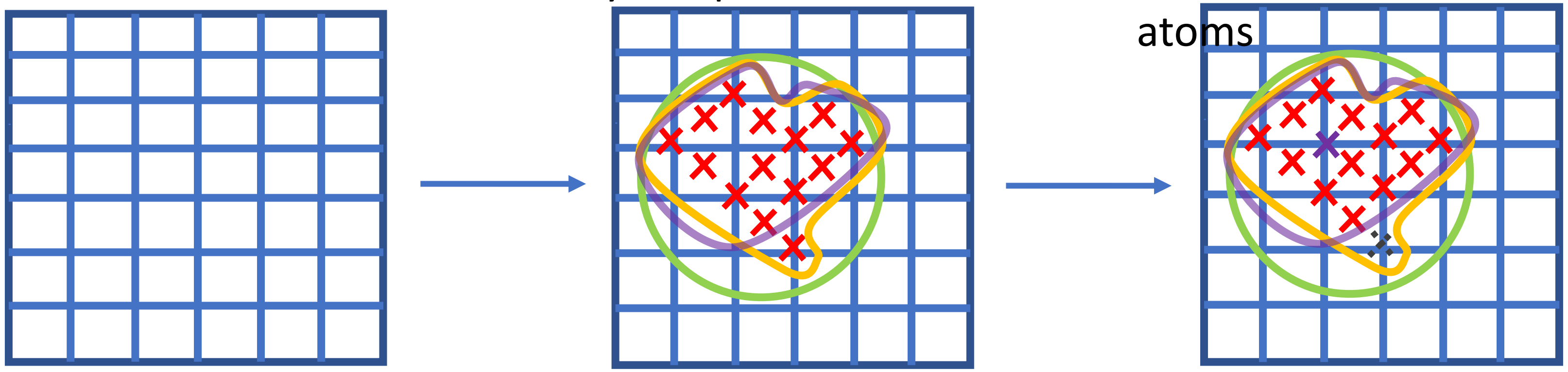
particle database

Random particle generator:

nxnxn empty lattice space

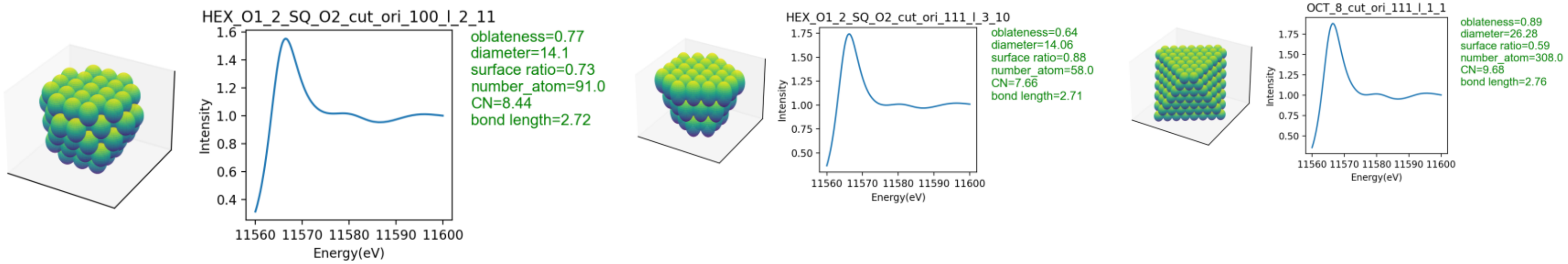
Randomly sample atoms on vertices

Fill holes using outmost atoms



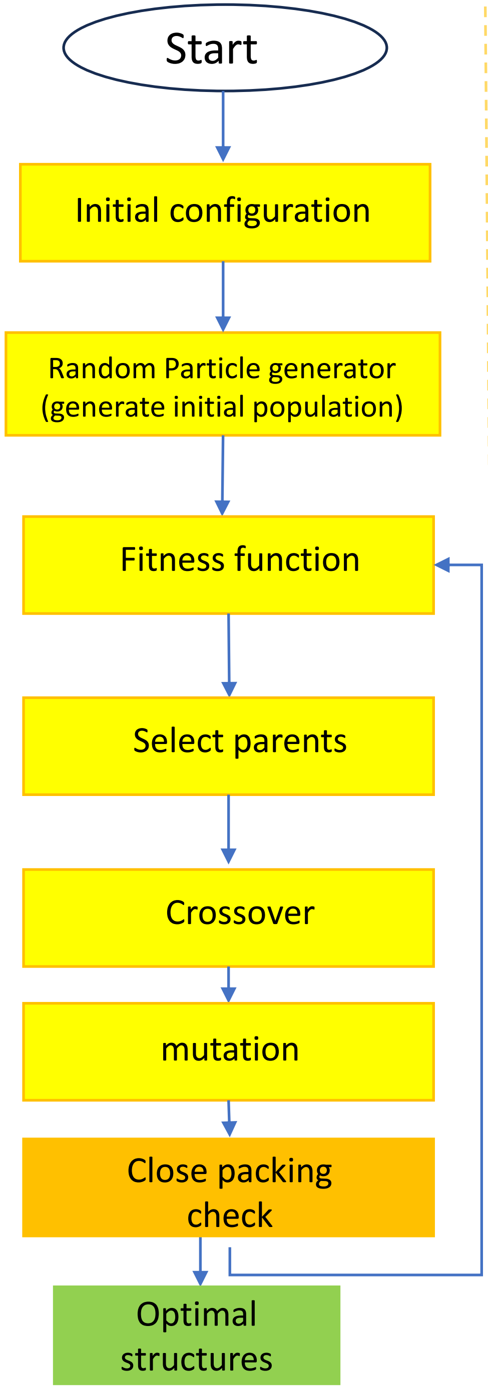
Constrains:

Bulk: $CN = 12$
Surface: $CN < 12$
Holes: $CN > 4$, no atom in the site



Regular particles

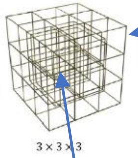
	CN1	CN2	CN3	CN4	diameter_2radius	diameter_pca	diameter_xy	MIAD	surface ratio	Departure from sphere(moment)	flattening_moment	flattening_pca	atom_number
OCT_2	4.00	1.00	0.00	0.00	3.77	6.53	3.77	2.89	1.00	0.000000	1.00	1.00	6.0
OCT_3	6.32	1.89	5.05	1.89	7.54	13.06	7.54	4.16	0.95	0.000000	0.00	1.00	19.0
OCT_4	7.64	2.59	8.73	3.82	11.31	18.24	11.31	5.49	0.86	0.000000	1.00	1.00	44.0
OCT_5	8.47	3.11	11.29	5.22	15.08	25.96	15.08	6.84	0.78	0.000000	0.00	1.00	85.0
OCT_6	9.04	3.49	13.15	6.25	18.85	29.81	18.85	8.19	0.70	0.000000	1.00	1.00	146.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
HEX_O1_2_SQ_O2_cut_ori_100_l_2	8.44	4.66	3.21	2.86	13.81	20.30	13.59	7.03	0.73	0.050589	0.77	0.57	91.0
HEX_O1_2_SQ_O2_cut_ori_100_l_3	7.89	4.11	3.06	2.57	14.26	19.68	13.59	6.70	0.81	0.109811	0.68	0.42	70.0
HEX_O1_2_SQ_O2_cut_ori_100_l_4	7.13	2.96	2.74	5.13	12.24	15.55	11.92	6.05	0.91	0.172209	0.61	0.35	46.0
HEX_O1_2_SQ_O2_cut_ori_100_l_5	5.76	2.88	2.08	3.84	11.39	11.47	11.31	5.30	1.00	0.244533	0.56	0.24	25.0
HEX_O1_2_SQ_O2_cut_ori_100_l_6	2.67	1.78	1.33	1.78	7.54	10.12	7.54	4.36	1.00	0.333333	0.50	0.00	9.0



❑ Initial configuration

- Descriptors
- Mutation rate
- Crossover rate
- Maximum number of atoms
- Population size

❑ Random particle generator



Big box to keep genome with the same size

Control shape(inner box:2x2x2)

❑ fitness function

Descriptors:$[CN_1, CN_2, CN_3, CN_4, D, N_{atoms} ...]$

$$Fitness = \max(abs(\frac{Descriptors_{particle}}{Descriptors_{true}} - 1))$$

Relative error

❑ Driving force

$$\min fitness(Descriptors)$$

❑ Select parents

- High probability to select individuals with small fitness value[3]

$$P = \frac{e^{-\lambda r}}{\Sigma_{r'=1}^{R} e^{-\lambda r'}}$$

Fitness ranking

0.1,0.2,0.1,0.3,0.6,0.5

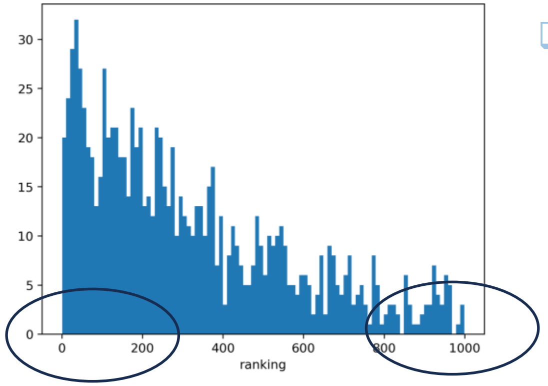
Ranking:

1,2,1,3,5,4

Where r is the ranking values, lambda controls the probability difference between the worst-ranking and best ranking

$$\lambda = \frac{ln10}{P_{max} - P_{min}}$$

The best ranking case was 10 times likely to choose be parents than worst-ranking

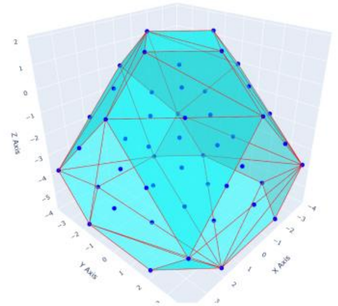


best rank, high probability

worst rank, low probability

❑ Close packing check

Convex hull

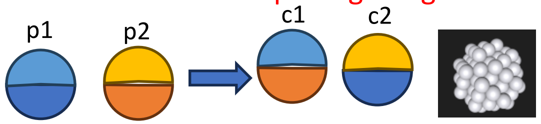


```
if random.uniform(0,1)<=prob[i]:
```

❑ Crossover

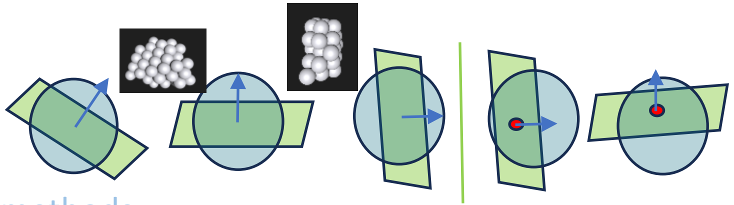
How should we keep the good genome?

Truncate and mix



❑ Mutation

truncation

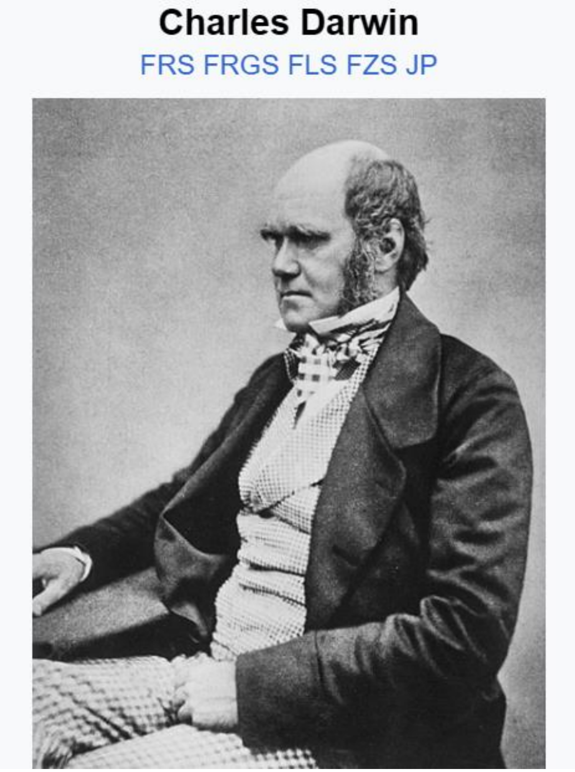


❑ Diversity check methods

- Elitism
- Crowding
- Niche

[3] N.S. Froemming, G. Henkelman J. Chem. Phys. 131, 234103 (2009)

Genetic algorithm

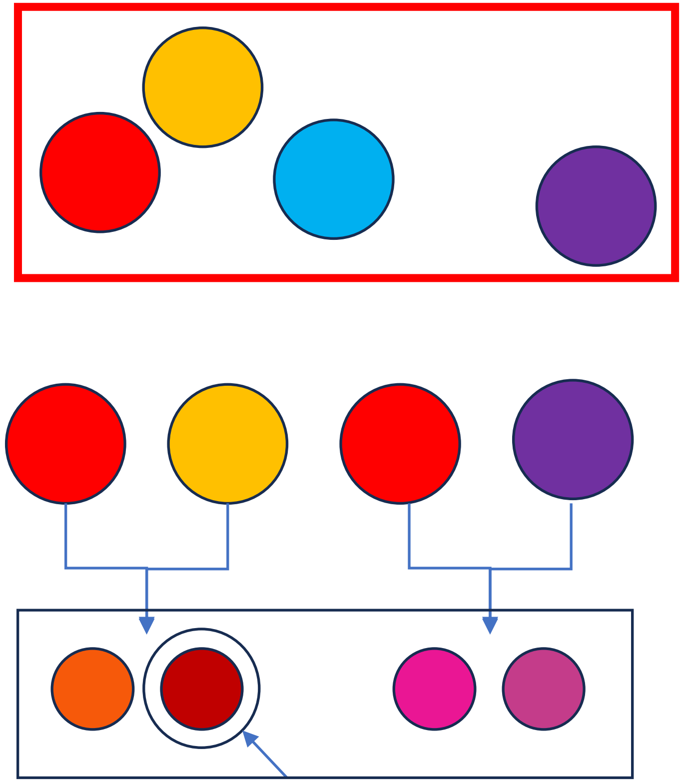


Before selection

Preference: red(driving force)

population

Parents(gene crossover)



More red!

Next generation

mutation

“*nature* *selection*”

Driving force

Statistics(good or bad)